

Measure-Theoretical Construction and Unbiased
MCMC Simulation of Brownian Motion

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Abstract

Brownian motion is a fundamental stochastic process that lies at the intersection of various classes of processes and with widespread applications across mathematical sciences. This work investigates it from its theoretical foundations to its numerical simulation. We begin with a rigorous development of stochastic processes within the framework of measure-theoretic probability and construct the Brownian motion through its finite-dimensional distributions. Next, we present the ε -strong algorithm introduced by Beskos et al., an iterative sampling method that delivers upper and lower bounding processes converging almost surely in both the supremum and L1 norms. Unlike standard discretization-based methods, this algorithm is, by construction, free of discretization bias. We start by providing an overview of its core structure (i.e., intersection layers) and providing pseudocode for each routine. We then provide derivations of all the distributions and sampling quantities that need to be computed. Finally, we provide a Python implementation of the algorithm, which can be used as a starting point to Monte Carlo estimate path-dependent functionals.

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Chapter 1

Introduction

Brownian motion is a fundamental stochastic process whose study has profoundly influenced both theory and applications across different scientific disciplines. In this chapter, we review its history, breadth of applications, and simulation challenges. Also, we will point to a Python implementation of the simulation algorithm.

1.1 Background and Motivation

It was first described by the botanist Robert Brown in 1827, who observed the random movement of pollen particles suspended in water [Bro28], but it wasn't until Albert Einstein's seminal work [Ein05] and Norbert Wiener's axiomatic construction [Wie23] that the motion of those particles was given a mathematical foundation: a continuous-time process with stationary, independent Gaussian increments. Today, Brownian motion transcends its origins in statistical physics. It provides the mathematical underpinnings for the Black-Scholes model [BS73] in quantitative finance, where log-prices of assets are modeled as drifted Brownian motion, and it serves as the usual “building block” in stochastic differential equations across numerous fields (e.g., signal processing in engineering or population genetics in biology).

1.2 Simulation Challenges and Implementation

Brownian motion generates continuous but nowhere differentiable paths that display fractal irregularity at every scale. As a result, important functionals such as the supremum of the process over a time interval, the hitting time to a given level, or the occupation time above a threshold depend on the entire trajectory and cannot be recovered from the terminal value alone. Therefore, accurately estimating such functionals requires simulating the continuous path itself.

The simplest way to simulate Brownian motion is to sample at equally spaced time points and linearly interpolate between them. This approach, while straightforward, introduces a discretization bias: extremes between grid points are missed, and hitting times are only approximated. Halving the step-size reduces the bias, but never eliminates it. For many applications, the remaining error can be unacceptable or very costly to control. Thus, when high precision is crucial, simulation methods that deliver unbiased estimators are needed.

Algorithms in this thesis are implemented in Python, and the full source code is available on GitHub:
<https://github.com/smauloni/epsilon-strong-BM>

Chapter 2

Stochastic Processes

In this chapter, we provide an introduction to the fundamental theory behind stochastic processes, discussing key concepts that will serve as the foundation for the next chapter, where we focus on Brownian motion. For an in-depth review of the measure theory needed for the discussion, the reader can see Appendix A.

2.1 Definition

In order to define stochastic processes, we first need to understand the spaces we are dealing with. We define two measurable spaces on which our objects will be based:

- Sample space : $(\Omega, \mathcal{F})$, on which probability measures P are defined.
- State space: $(S, \mathcal{S})$, on which the process takes values.

In particular, in our discussion we will consider $S = \mathbb{R}^d$ and $\mathcal{S} = \mathcal{B}(\mathbb{R}^d)$, where $\mathcal{B}(U)$ denotes the Borel σ -algebra of a space U .

Definition 2.1. A *stochastic process* is a collection of random variables $X = \{X_t : 0 \leq t < \infty\}$ taking values in $(S, \mathcal{S})$. Fixing $\omega \in \Omega$, the map $t \mapsto X_t(\omega)$ $\{t \geq 0\}$ is called *sample path (realization, trajectory)* of the process X associated with ω .

Often, the index t is referred to as time. This is convenient since defining past, present, and future becomes intuitive.

Definition 2.2. A stochastic process is *measurable* if the map

$$(t, \omega) \mapsto X_t(\omega) : ([0, \infty] \times \Omega, \mathcal{B}([0, \infty] \otimes \mathcal{F})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}^d))$$

is measurable. In other words, this means that $\forall A \in \mathcal{B}(\mathbb{R}^d)$, the preimage of A lies in the product σ -algebra of $[0, \infty] \times \Omega$.

Here, we will only treat measurable stochastic processes.

By Fubini's theorem, trajectories of such stochastic processes $(t \mapsto X_t(\omega))$ are Borel-measurable. Moreover, if $\forall t, E(X_t)$ is finite, then $m(t) := E(X_t)$ is Borel-measurable as well.

2.2 Filtrations

To make sense of the concept of time for stochastic processes, and keep information about past, present, and future at time t , it is necessary to consider filtrations.

Definition 2.3. A *filtration* is a nondecreasing family $\{\mathcal{F}_t, t \geq 0\}$ of sub- σ -algebras of $\mathcal{F}$:

$\forall s < t, \mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$. Set $\mathcal{F}_\infty := \sigma(\cup_{t \geq 0} \mathcal{F}_t)$

Given a stochastic process, we commonly choose the filtration *generated* by the process itself: $\mathcal{F}_t^X := \sigma(X_s : 0 \leq s \leq t)$, which is the smallest σ -algebra for which X_s is measurable $\forall s \in [0, t]$.

Given our definition, $A \in \mathcal{F}_t^X$ implies that we know whether A has been observed or not. Thus, we can define $\mathcal{F}_{t-} := \sigma(\cup_{s \leq t} \mathcal{F}_s)$ (σ -algebra of events strictly prior to t) and $\mathcal{F}_{t+} := \sigma(\cap_{\epsilon \geq 0} \mathcal{F}_{t+\epsilon})$ (σ -algebra of events immediately after t). We also define $\mathcal{F}_{0-} := \mathcal{F}_0$

We say that the filtration $\mathcal{F}_t$ is *right(left)-continuous* if $\mathcal{F}_{t+} = \mathcal{F}_t$ ($\mathcal{F}_{t-} = \mathcal{F}_t$) $\forall t \geq 0$.

Example. Consider the process X_t , defined as:

$$X_t = \begin{cases} 0, & \text{if } t < T \\ 1, & \text{if } t \geq T \end{cases}$$

where T is a random variable representing the time at which X_t jumps from 0 to 1. Let $T \sim Unif[0, 1]$.

- Sample Space (Ω): The possible values of T , so $\Omega = [0, 1]$.
- State Space: The possible values of X_t , so $S = \{0, 1\}$.

$\mathcal{F}_t$ is the σ -algebra containing all events that can be determined from observing whether T (the jump time) is less than or greater than t .

1. For $t = 0$: We know nothing about T , so: $\mathcal{F}_0 = \{\emptyset, \Omega, \{T = 0\}, \{T > 0\}\}$.
2. For $t > 0$: We can determine if $T \leq t$. Thus, the filtration expands to: $\mathcal{F}_t = \sigma(X_s : 0 \leq s \leq t)$.
3. For $t \geq 1$: T is revealed since $T \leq 1$ is true. Thus: $\mathcal{F}_1 = \mathcal{F}_\infty = \{\emptyset, \Omega, \text{all measurable subsets of } [0, 1]\}$.

Definition 2.4. A process X is *adapted* to the filtration $\{\mathcal{F}_t\}$ if $\forall t \geq 0$, X_t is a $\mathcal{F}_t$ -measurable random variable. Measurability is denoted by $X_t \in \mathcal{F}_t$.

Trivially, X_t is adapted to $\mathcal{F}_t^X$. In addition, if Y_t is a modification of X_t (see next section), it is adapted to $\mathcal{F}_t^X$ as well, provided $\mathcal{F}_0$ contains all P -negligible sets in $\mathcal{F}$.

2.3 Modifications

It is often useful to consider different stochastic processes rather than the one we are interested in studying in order to grasp information about them. We provide some definitions of transformations.

Definition 2.5. Consider two stochastic processes Y_t and X_t defined on the same probability space $(\Omega, \mathcal{F}, P)$. Then:

- (i) Y is a *modification* of X if $\forall t \geq 0, P[X_t = Y_t] = 1$;
- (ii) X and Y have the *same finite-dimensional distributions* if for $n \geq 1$,
 $0 \leq t_0 < t_1 < \dots < t_n < \infty$, and $A \in \mathcal{B}(\mathbb{R}^d)$, then $P[(X_{t_1}, \dots, X_{t_n}) \in A] = P[(Y_{t_1}, \dots, Y_{t_n}) \in A]$
- (iii) X and Y are *indistinguishable* if $P[X_t = Y_t \forall t \geq 0] = 1$.

Note: (iii) $\implies$ (i) $\implies$ (ii).

We can adapt (ii) to allow for different probability spaces:

Definition 2.6. X and Y defined on $(\Omega, \mathcal{F}, P)$ and $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{P})$ respectively and sharing the same state space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ have the *same finite-dimensional distributions* if for $n \geq 1$, $0 \leq t_0 < t_1 < \dots < t_n < \infty$, and $A \in \mathcal{B}(\mathbb{R}^d)$, then $P[(X_{t_1}, \dots, X_{t_n}) \in A] = \tilde{P}[(Y_{t_1}, \dots, Y_{t_n}) \in A]$.

One should note that (ii) is important since many stochastic processes are defined by their finite-dimensional distribution (e.g., Gaussian processes), independently of the probability measure associated.

Chapter 3

Brownian Motion

Brownian motion plays a fundamental role in probability theory and stochastic processes. In this chapter, we construct it and explore its properties, which make it an essential model for randomness, serving as the foundation for stochastic calculus.

3.1 Definition

The standard, one-dimensional Brownian motion is a continuous, adapted process $B = \{B_t, \mathcal{F}_t; 0 \leq t < \infty\}$ defined on a probability space $(\Omega, \mathcal{F}, P)$ and such that:

- (i) $B_0 = 0$ a.s.;
- (ii) for $0 \leq s < t$, $(B_t - B_s) \sim N(0, t - s)$ and is independent of $\mathcal{F}_s$.

We can also restrict it on $[0, T]$ to be $B = \{B_t, \mathcal{F}_t; 0 \leq t \leq T\}$, with $T > 0$.

By (ii), we say that B has stationary, independent increments. Also, B is square-integrable.

Note that even though the filtration $\{\mathcal{F}_t\}$ is part of the definition, $B = \{B_t; 0 \leq t < \infty\}$ satisfying (i) and (ii) is still a Brownian motion if we include the generated Filtration $\{\mathcal{F}_t\}^B$. However, it is often interesting and necessary to work with a filtration larger than $\{\mathcal{F}_t\}^B$. The existence of solutions to stochastic differential equations is a major motivation.

Of course, we can extend the definition to the d -dimensional space:

Definition 3.1. Let d be a positive integer and μ a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. Let $B = \{B_t, \mathcal{F}_t; t \geq 0\}$ be a continuous, adapted process with values in $\mathbb{R}^d$, defined on some probability space $(\Omega, \mathcal{F}, P)$. This process is called a *d -dimensional Brownian motion with initial distribution μ* , if

- (i) $P[B_0 \in \Gamma] = \mu(\Gamma)$, $\forall \Gamma \in \mathcal{B}(\mathbb{R}^d)$;
- (ii) for $0 \leq s < t$, the increment $B_t - B_s$ is independent of $\mathcal{F}_s$ and is normally distributed with mean zero and covariance matrix equal to $(t - s)I_d$, where I_d is the $(d \times d)$ identity matrix.

3.2 Properties

Brownian motion is a remarkably rich and fundamental example of a stochastic process. It exhibits a wide range of properties that are often studied individually in broader classes of processes. This makes

it a central object of study in probability theory. We now present some of the main ones. Of course, this list does not intend to be comprehensive of all properties the Brownian motion has, but rather to provide some intuition for how it behaves and what its trajectories "look like".

3.2.1 Markovianity

Definition 3.2. Let μ be a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. An adapted, d -dimensional process $X = \{X_t, \mathcal{F}_t; t \geq 0\}$ on a probability space $(\Omega, \mathcal{F}, P^\mu)$ is said to be a *Markov process with initial distribution* μ if:

- (i) $P^\mu[X_0 \in \Gamma] = \mu(\Gamma), \forall \Gamma \in \mathcal{B}(\mathbb{R}^d),$
- (ii) $\forall s, t \geq 0, \forall \Gamma \in \mathcal{B}(\mathbb{R}^d), P^\mu[X_{t+s} \in \Gamma | \mathcal{F}_s] = P^\mu[X_{t+s} \in \Gamma | X_s], \quad P\text{-a.s.}$

This means that we are only interested in the present, not the past, when computing probabilities for future values of X .

Theorem 3.3. *A d -dimensional Brownian motion is a Markov process.*

3.2.2 Martingale

Definition 3.4. A process $X = \{X_t, \mathcal{F}_t; t \geq 0\}$ is a *martingale* with respect to a filtration $\{\mathcal{F}_t\}$ if:

- (i) It is adapted (see 2.4);
- (ii) $E|X_t| < +\infty$ for all $t \geq 0$;
- (iii) $E[X_t | \mathcal{F}_s] = X_s$, almost surely, for all $0 \leq s \leq t$.

This means that, given the present, our best guess for the future is simply the present; there is no predictable trend up or down based on the past.

Theorem 3.5. *The Brownian motion is a martingale.*

Proof. Let $\{B_t\}$ be a standard Brownian motion. Then, by the Markov property:

$$E[B_s | \mathcal{F}_s^+] = E[B_t - B_s | \mathcal{F}_s^+] + B_s = E[B_t - B_s] + B_s = B_s.$$

□

3.2.3 Scaling

If B_t is a standard Brownian motion, then for any constant $c > 0$, the rescaled process

$$\tilde{B}_t := \frac{1}{\sqrt{c}} B_{ct}$$

is also a standard Brownian motion.

This property tells us that zooming in or out in time and space, the motion still "looks" like Brownian motion, since the randomness has no natural scale.

3.3 Construction

There are several ways to construct the Brownian motion. Here, we focus on the "direct approach", consisting of writing down what the finite-dimensional distributions of this process (based on the stationarity, independence, and normality of its increments) must be, and then constructing a probability measure and a process on an appropriate measurable space in such a way that we obtain the prescribed finite-dimensional distributions. This direct approach is the one most often used to construct a Markov process, but it is lengthy and technical. ¹

The construction consists of two elements: the Consistency theorem and the Kolmogorov-Centsov theorem.

3.3.1 The Consistency Theorem

It will turn out to be useful to use cylinder sets, which can be thought of as "discretizations" of trajectories of a stochastic process. These will allow us to explicitly define the finite-dimensional distribution we are searching for.

Definition 3.6. Let $\mathbb{R}^{[0,\infty)}$ denote the set of all real-valued functions on $[0,\infty)$. An *n-dimensional cylinder set* in $\mathbb{R}^{[0,\infty)}$ is of the form

$$C := \{\omega \in \mathbb{R}^{[0,\infty)} \mid (\omega(t_1), \dots, \omega(t_n)) \in A\},$$

where $t_i \in [0, \infty)$ for $i = 1, \dots, n$, and $A \in \mathcal{B}(\mathbb{R}^n)$. Let $\mathcal{C}$ denote the algebra of all such cylinder sets, and let $\mathcal{B}(\mathbb{R}^{[0,\infty)})$ be the smallest σ -algebra containing $\mathcal{C}$.

Definition 3.7. Let $\mathcal{T}$ be the set of finite sequences $\mathbf{t} = (t_1, \dots, t_n)$ of distinct, nonnegative numbers, where the length of these sequences is $n \in \mathbb{N}_+$.

For each $\mathbf{t}$ of length n , suppose we have a probability measure $Q_{\mathbf{t}}$ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Then the collection $\{Q_{\mathbf{t}}\}_{\mathbf{t} \in \mathcal{T}}$ is called a *family of finite-dimensional distributions*.

This family is *consistent* if:

- (i) $Q_{t_{\pi(1)}, \dots, t_{\pi(n)}}(A_{\pi(1)} \times \dots \times A_{\pi(n)}) = Q_{t_1, \dots, t_n}(A_1 \times \dots \times A_n)$ for every permutation π .
- (ii) For $n \geq 1$, $\mathbf{t} = (t_1, \dots, t_{n-1}, t_n)$, $\mathbf{s} = (t_1, \dots, t_{n-1})$, and $A \in \mathcal{B}(\mathbb{R}^{n-1})$, then $Q_{\mathbf{t}}(A \times \mathbb{R}) = Q_{\mathbf{s}}(A)$ for $A \in \mathcal{B}(\mathbb{R}^{n-1})$

If there exists a probability measure P on $(\mathbb{R}^{[0,\infty)}, \mathcal{B}(\mathbb{R}^{[0,\infty)}))$, we can define a family of finite-dimensional distributions via

$$Q_{\mathbf{t}}(A) = P \left[\omega \in \mathbb{R}^{[0,\infty)} \mid (\omega(t_1), \dots, \omega(t_n)) \in A \right], \quad (3.1)$$

¹An alternative, not treated here, exploits the Gaussian property of this process and is based on Hilbert space theory, which is similar to Wiener's original work, later modified by Lévy and Ciesielski.

where $A \in \mathcal{B}(\mathbb{R}^n)$ and $\mathbf{t} = (t_1, \dots, t_n) \in \mathcal{T}$.

This family is always consistent. We are particularly interested in the converse: can we construct a probability measure P from a given family of finite-dimensional distributions? This will enable us to construct a probability measure P from the finite-dimensional distributions of the Brownian motion.

First, we define regular elements of the sigma-algebra and state that every element is regular, indeed.

Definition 3.8. Let Q be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. We say that $A \in \mathcal{B}(\mathbb{R}^n)$ is *regular* if for every probability measure Q on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ and for every $\epsilon > 0$, there is a closed set F and an open set G such that $F \subseteq A \subseteq G$ and $Q(G \setminus F) < \epsilon$.

Proposition 3.9. A set in $\mathcal{B}(\mathbb{R}^n)$ is regular, since the collection of regular sets is a σ -algebra containing all closed sets.

Proof. The reader can see [KS91], solution to Problem 2.4. $\square$

Theorem 3.10 (Daniell (1918), Kolmogorov (1933)). *Let $\{Q_{\mathbf{t}}\}$ be a consistent family of finite-dimensional distributions. Then there exists a probability measure P on $(\mathbb{R}^{[0,\infty)}, \mathcal{B}(\mathbb{R}^{[0,\infty)}))$ such that (3.1) holds $\forall \mathbf{t} \in T$.*

Proof. We proceed as follows: we begin by establishing the countable additivity of Q . Then, by the Hahn–Kolmogorov–Carathéodory extension theorem (A.7), we assert the existence of the measure extension P of Q to $\mathcal{B}(\mathbb{R}^{[0,\infty)})$.

We start by defining $\{B_k\}_{k=1}^\infty$ a sequence of disjoint sets in $\mathcal{C}$ and such that $B := \bigcup_{k=1}^\infty B_k \in \mathcal{C}$. Then, let $C_m = B \setminus \bigcup_{k=1}^m B_k$. Then, $Q(B) = Q(C_m) + \sum_{k=1}^m Q(B_k)$. We show countable additivity if $\lim_{m \rightarrow \infty} Q(C_m) = 0$.

Note $Q(C_m) = Q(C_{m+1}) + Q(B_{m+1}) \geq Q(C_{m+1})$. Thus, the limit exists.

By contradiction, assume that the limit is equal to $\epsilon > 0$. Note that $\bigcap_{m=1}^\infty C_m = \emptyset$.

Starting from $\{C_m\}_{m=1}^\infty$ we construct another sequence $\{D_m\}_{m=1}^\infty$ such that $D_1 \supseteq D_2 \supseteq \dots$, $\bigcap_{m=1}^\infty D_m = \emptyset$, and $\lim_{m \rightarrow \infty} Q(D_m) = \epsilon > 0$. Moreover, each $D_m = \{\omega \in \mathbb{R}^{[0,\infty)}; (\omega(t_1), \dots, \omega(t_m)) \in A_m\}$, with $A \in \mathcal{B}(\mathbb{R}^m)$, and each finite $\mathbf{t}_m := (t_1, \dots, t_m) \in T$ is an extension of $\mathbf{t}_{m-1} := (t_1, \dots, t_{m-1}) \in T$ for $m \geq 2$.

To construct such $\{D_m\}_{m=1}^\infty$, we start by noting that

$$C_k = \{\omega \in \mathbb{R}^{[0,\infty)}; (\omega(t_1), \dots, \omega(t_{m_k})) \in A_{m_k}\}, \quad A_{m_k} \in \mathcal{B}(\mathbb{R}^{m_k}), \quad (t_1, \dots, t_{m_k}) \in T.$$

Since $C_{k+1} \subseteq C_k$, we can choose a representation such that $\mathbf{t}_{m_{k+1}}$ is an extension of $\mathbf{t}_{m_k}$ and $A_{m_{k+1}} \subseteq A_{m_k} \times \mathbb{R}^{m_{k+1}-m_k}$. Define

$$D_1 = \{\omega; \omega(t_1) \in \mathbb{R}\}, \dots, D_{m_t-1} = \{\omega; (\omega(t_1), \dots, \omega(t_{m_t-1})) \in \mathbb{R}^{m_t-1}\}$$

and $D_{m_1} = C_1$. Also,

$$D_{m_1+1} = \{\omega; (\omega(t_1), \dots, \omega(t_{m_1}), \omega(t_{m_1+1})) \in A_{m_1} \times \mathbb{R}\}, \dots,$$

$$D_{m_2-1} = \{\omega; (\omega(t_1), \dots, \omega(t_{m_1}), \omega(t_{m_1+1}), \dots, \omega(t_{m_2-1})) \in A_{m_1} \times \mathbb{R}^{m_2-m_1-1}\},$$

and $D_{m_2} = C_2$. Continuing this process, we get that by construction $\bigcap_{m=1}^{\infty} D_m = \bigcap_{m=1}^{\infty} C_m = \emptyset$.

By 3.9, for every m there exists a closed $F_m \subseteq A_m$ such that $Q_{\mathbf{t}_m}(A_m \setminus F_m) < \frac{\epsilon}{2^m}$. Intersecting F_m with a sufficiently large closed sphere centered at the origin, we obtain a compact K_m such that, with $E_m := \{\omega \in \mathbb{R}^{[0,\infty)}; (\omega(t_1), \dots, \omega(t_m)) \in K_m\}$, we have $E_m \subseteq D_m$ and

$$Q(D_m \setminus E_m) = Q_{\mathbf{t}_m}(A_m \setminus K_m) < \frac{\epsilon}{2^m}.$$

Since $\{E_m\}$ might not be nonincreasing, we define $\tilde{E}_m = \bigcap_{k=1}^m E_k$, and we have

$$\tilde{E}_m = \{\omega \in \mathbb{R}^{[0,\infty)}; (\omega(t_1), \dots, \omega(t_m)) \in \tilde{K}_m\},$$

where

$$\tilde{K}_m = (K_1 \times \mathbb{R}^{m-1}) \cap (K_2 \times \mathbb{R}^{m-2}) \cap \dots \cap (K_{m-1} \times \mathbb{R}),$$

which is a compact set. We can bound $Q_{\mathbf{t}_m}(\tilde{K}_m)$ away from zero now, since

$$\begin{aligned} Q_{\mathbf{t}_m}(\tilde{K}_m) &= Q(\tilde{E}_m) = Q(D_m) - Q(D_m \setminus \tilde{E}_m) \\ &= Q(D_m) - Q(\bigcup_{k=1}^m (D_m \setminus E_k)) \\ &\geq Q(D_m) - Q(\bigcup_{k=1}^m (D_k \setminus E_k)) \geq \epsilon - \sum_{k=1}^m \frac{\epsilon}{2^k} > 0. \end{aligned}$$

Thus, for every m , $\tilde{K}_m$ is nonempty. Hence, we can choose $(x_1^{(m)}, \dots, x_m^{(m)}) \in \tilde{K}_m$. Since it is contained in the compact set $\tilde{K}_1$, the sequence $\{x_1^{(m)}\}_{m=1}^{\infty}$ must have a convergent subsequence $\{x_1^{(m_k)}\}_{k=1}^{\infty}$ with limit x_1 . However, $\{x_1^{(m_k)}, x_2^{(m_k)}\}_{k=2}^{\infty}$ is contained in $\tilde{K}_2$ and has a convergent subsequence. Iterating the process, we can construct a sequence $(x_1, x_2, \dots) \in \mathbb{R} \times \mathbb{R} \times \dots$, such that $(x_1, \dots, x_m) \in \tilde{K}_m \forall m$. Therefore,

$$S = \{\omega \in \mathbb{R}^{[0,\infty)} : \omega(t_i) = x_i, i = 1, 2, \dots\} \subseteq \tilde{E}_m \implies S \subseteq D_m \forall m.$$

This bring a contradiction with $\bigcap_{m=1}^{\infty} D_m = \emptyset$, implying that $\lim_{m \rightarrow \infty} Q(C_m) = 0$. $\square$

Now we aim to construct a probability measure P on $(\Omega, \mathcal{F}) := (\mathbb{R}^{[0,\infty)}, \mathcal{B}(\mathbb{R}^{[0,\infty)}))$ so that the coordinate mapping process $B = \{B_t, \mathcal{F}_t^B; 0 \leq t < \infty\}$ defined by $B_t(\omega) := \omega(t)$, is almost a standard, one-dimensional Brownian motion under P .

We say "almost" because we concentrate on the finite-dimensional distributions, leaving aside the requirement of sample path continuity for the moment. We will prove that in the next section.

We see that whenever $0 = s_0 < s_1 < s_2 < \dots < s_n$, the cumulative distribution function for $(B_{s_1}, \dots, B_{s_n})$ must be

$$\begin{aligned} F_{(s_1, \dots, s_n)}(x_1, \dots, x_n) &= \\ &= \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \dots \int_{-\infty}^{x_n} p(s_1; 0, y_1) p(s_2 - s_1; y_1, y_2) \dots p(s_n - s_{n-1}; y_{n-1}, y_n) dy_n \dots dy_2 dy_1, \end{aligned}$$

for $(x_1, \dots, x_n) \in \mathbb{R}^n$, where p is the Gaussian kernel $p(t; x, y) := \frac{1}{\sqrt{2\pi t}} e^{-(x-y)^2/2t}$, $t > 0$, $x, y \in \mathbb{R}$.

One can verify that this is equivalent to the statement that the increments $\{B_{s_j} - B_{s_{j-1}}\}_{j=1}^n$ are independent, and $B_{s_j} - B_{s_{j-1}}$ is normally distributed with mean zero and variance $s_j - s_{j-1}$.

Now let $\mathbf{t} = (t_1, \dots, t_n)$, with t_j distinct. Let $(B_{t_1} \dots B_{t_n})$ have distributions as above. For $A \in \mathcal{B}(\mathbb{R}^n)$, let $Q_{\mathbf{t}}(A)$ be the probability under that distribution that $(B_{t_1} \dots B_{t_n})$ is in A . This defines a consistent family of finite-dimensional distributions $\{Q_{\mathbf{t}}\}_{\mathbf{t} \in T}$.

Corollary 3.11. There exists a probability measure P on $(\mathbb{R}^{[0, \infty)}, \mathcal{B}(\mathbb{R}^{[0, \infty)}))$, under which the coordinate mapping process

$$B_t(\omega) = \omega(t); \quad \omega \in \mathbb{R}^{[0, \infty)}, t \geq 0$$

has stationary, independent increments. For $0 \leq s < t$, an increment $B_t - B_s$ is normally distributed with mean zero and variance $t - s$.

3.3.2 The Kolmogorov-Čentsov Theorem

In principle, the construction of the Brownian motion is complete. However, the process we have built is on $\mathbb{R}^{[0, \infty)}$, instead we need it to be in $C[0, \infty)$ (i.e., the space of continuous functions defined on the nonnegative half-line).

What we want to do is prove that P in (3.11) assigns probability one to $C[0, \infty)$. The issue with this is that $C[0, \infty)$ is not in the σ -algebra $\mathcal{B}(\mathbb{R}^{[0, \infty)})$. Thus, we can exploit a property announced before, that is, to use a larger σ -algebra which contains $\mathcal{B}(\mathbb{R}^{[0, \infty)})$. In doing so, we should be aware that it is not possible to have a function in $\mathbb{R}^{[0, \infty)}$ by specifying its values at only countably many coordinates, in contrast with $C[0, \infty)$. Thus, we want to take a different approach: to construct a continuous modification of the coordinate mapping process we defined in (3.11).

Theorem 3.12 (Kolmogorov-Čentsov (1956)). *Suppose the process $X = \{X_t; 0 \leq t < \infty\}$ on a probability space $(\Omega, \mathcal{F}, P)$ satisfies*

$$E(|X_t - X_s|^\alpha) \leq C|t - s|^{1+\beta}, \quad s, t \in [0, T]$$

for positive constants α, β, C . Then, there exists a continuous modification $\tilde{X} = \{\tilde{X}_t; 0 \leq t < \infty\}$, which is locally γ -Hölder-continuous $\forall \gamma \in (0, \frac{\beta}{\alpha})$, i.e.

$$P \left[\omega; \sup_{\substack{0 < t-s < h(\omega) \\ s, t \in [0, T]}} \frac{|\tilde{X}_t(\omega) - \tilde{X}_s(\omega)|}{|t-s|^\gamma} \leq \delta \right] = 1,$$

with $h(\omega)$ an a.s. positive random variable and $\delta > 0$ a constant.

Proof. To prove this theorem, we rely heavily on the Markov inequality. For simplicity, fix $T = 1$. For $\epsilon > 0$ we have

$$P(|X_t - X_s| \geq \epsilon) \leq \frac{E(|X_t - X_s|^\alpha)}{\epsilon^\alpha} \leq C\epsilon^{-\alpha}|t-s|^{1+\beta}.$$

Thus, $X_s \xrightarrow{P} X_t$ as $s \rightarrow t$. Setting $t = \frac{k}{2^n}$, $s = \frac{k-1}{2^n}$, and $\epsilon = 2^{-\gamma n}$, for $\gamma \in (0, \frac{\beta}{\alpha})$, we obtain

$$P(|X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}) \leq C2^{-n(1+\beta-\alpha\gamma)}.$$

Thus,

$$P \left(\max_{1 \leq k \leq 2^n} |X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n} \right) \leq P \left(\bigcup_{k=1}^{2^n} |X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n} \right) \leq C2^{-m(\beta-\alpha\gamma)}.$$

Note that $C2^{-m(\beta-\alpha\gamma)}$ is a general term of a convergent series. By the Borel-Cantelli lemma (A.2.3), $\exists \Omega^* \in \mathcal{F}$ with $P(\Omega^*) = 1$ and such that $\forall \omega \in \Omega^*$,

$$\max_{1 \leq k \leq 2^n} |X_{\frac{k}{2^n}}(\omega) - X_{\frac{k-1}{2^n}}(\omega)| < 2^{-\gamma n} \quad \forall n \geq n^*(\omega), \quad (3.2)$$

with $n^*(\omega)$ a $\mathbb{N}_+$ -valued random variable.

Now for $n \geq 1$, we partition $[0, 1]$ with $D_n = \{(k/2^n); k = 0, 1, \dots, 2^n\}$. Let also $D := \bigcup_{n=1}^{\infty} D_n$ be the set of dyadic rationals in the interval $[0, 1]$. Now we would like to fix $\omega \in \Omega^*$, $n \geq n^*(\omega)$ and show that

$$|X_t(\omega) - X_s(\omega)| \leq 2 \sum_{j=n+1}^m 2^{-\gamma j}, \quad \forall t, s \in D_m, \quad 0 \leq t-s \leq 2^{-n}. \quad (3.3)$$

For $m = n+1$, we can just set $t = \frac{k}{2^m}$, $s = \frac{k-1}{2^m}$ and (3.3) would follow from (3.2). Suppose now that this is valid for $m = n+1, \dots, M-1$. Take $s < t, s, t \in D_M$ and consider $t^1 = \max\{u \in D_{m-1} : u \leq t\}$ and $s^1 = \min\{u \in D_{m-1} : u \geq s\}$. We get $s \leq s^1 \leq t^1 \leq t$, $s^1 - s \leq 2^{-M}$, and $t - t^1 \leq 2^{-M}$. Then, by (3.2) we have $|X_{s^1}(\omega) - X_s(\omega)| \leq 2^{-\gamma M}$ and $|X_t(\omega) - X_{t^1}(\omega)| \leq 2^{-\gamma M}$ and by (3.3) with $m = M-1$, $|X_{t^1}(\omega) - X_{s^1}(\omega)| \leq 2 \sum_{j=n+1}^{M-1} 2^{-\gamma j}$. We obtain (3.3) for $m = M$.

Now we can show that $\{X_t(\omega) : t \in D\}$ is uniformly continuous in $t \forall \omega \in \Omega^*$. For any $s, t \in D$ such that $0 < t - s < h(\omega) := 2^{-n^*(\omega)}$, select $n \geq n^*(\omega)$ such that $2^{-(n+1)} \leq t - s \leq 2^{-n}$. From (3.3) we have

$$|X_t(\omega) - X_s(\omega)| \leq 2 \sum_{j=n+1}^{\infty} 2^{-\gamma j} \leq \delta |t - s|^\gamma, \quad 0 < t - s < h(\omega), \quad \delta = \frac{2}{1 - 2^{-\gamma}}. \quad (3.4)$$

This proves uniform continuity.

At this point, we can define $\tilde{X}$: $\tilde{X}_t(\omega) = 0$ if $\omega \notin \Omega^*, t \in [0, 1]$ and $\tilde{X}_t(\omega) = X_t(\omega)$ if $\omega \in \Omega^*, t \in D$.

If instead $\omega \in \Omega^*, t \in [0, 1] \cap D^c$, choose $\{s_n\}_{n=1}^{\infty} \subseteq D, s_n \rightarrow t$. Since the sequence is Cauchy, uniform continuity implies that $\{X_{s_n}(\omega)\}_{n=1}^{\infty}$ has a limit dependent on t but not in $\{s_n\}$. Set $\tilde{X}_t(\omega) = \lim_{s_n \rightarrow t} X_{s_n}(\omega)$.

The resulting process $\tilde{X}$ is continuous by (3.4). Hence, the condition in the statement is verified.

Moreover, $\tilde{X}_t = X_t$ a.s. for $t \in D$. Instead, for $t \in [0, 1] \cap D^c$ and $\{s_n\}_{n=1}^{\infty} \subseteq D$ with $s_n \rightarrow t$, we have $X_{s_n} \xrightarrow{P} X_t$ and $X_{s_n} \rightarrow \tilde{X}_t$ almost surely. This implies $\tilde{X}_t = X_t$ almost surely. $\square$

In the interest of brevity, the following proposition is given without proof. For reference, the reader might want to see [KS91], Problem 2.10.

Proposition 3.13. If for $s < t$, $(B_t - B_s) \sim N(0, (t - s))$, then for each $n \in \mathbb{N}_+$, there is a positive constant C_n for which

$$E(|B_t - B_s|^{2n}) = C_n |t - s|^n.$$

Corollary 3.14. There exist a probability measure P on $(\mathbb{R}^{[0, \infty)}, \mathcal{B}(\mathbb{R}^{[0, \infty)}))$ and a stochastic process $W = \{W_t, \mathcal{F}_t^W; t \geq 0\}$ on the same space such that under P , W is a Brownian motion.

Proof. By (3.12) and (3.13), for every $T > 0$ there exists a modification W^T of B in (3.11) such that W^T is continuous on $[0, T]$. Now let $\Omega_T := \{\omega : W_t^T(\omega) = B_t(\omega) \forall t \in [0, T] \cap \mathbb{Q}\}$. We have $P(\Omega_T) = 1$.

On $\tilde{\Omega} := \bigcap_{T=1}^{\infty} \Omega_T$, we have for $T_1, T_2 \in \mathbb{N}_+$, $W_t^{T_1}(\omega) = W_t^{T_2}(\omega)$ for every $t \in [0, T_1 \wedge T_2] \cap \mathbb{Q}$ and $\omega \in \tilde{\Omega}$.

We just define $W_t(\omega)$ as the common value of the two. If $\omega \notin \tilde{\Omega}$, then we set $W_t(\omega) = 0 \forall t \geq 0$. $\square$

Remark. As a consequence of the previous results, we highlight that the Brownian sample paths $\{W_t(\omega) : 0 \leq t \leq \infty\}$ are locally Hölder-continuous with exponent $\gamma \in (0, \frac{1}{2})$ P -almost everywhere.

3.3.3 Brownian Bridge

Crucial for the next chapter, we now define the Brownian bridge, which can be seen as just a portion of Brownian motion conditioned on both ends.

Definition 3.15. Given a standard Brownian motion B_t , the process defined as

$$X_t := B_t - \frac{t}{T} B_T, \quad t \in [0, T];$$

is a Brownian bridge on $[0, T]$, conditioned to start and end at 0.

Chapter 4

Simulation of Brownian Paths

This chapter presents the ε -strong simulation algorithm for the Brownian motion, as introduced by Beskos et al. (2012) [BPR11]. The key idea is to construct upper and lower bounding processes that almost surely enclose the Brownian path, enabling exact simulation with a uniform error bound ε . This is achieved without relying on time discretization, which typically introduces approximation errors. Instead, the algorithm iteratively builds piecewise constant paths using information about the minimum and maximum values of the Brownian bridge over subintervals.

The upper and lower bounds, denoted $X^{\uparrow(n)}$ and $X^{\downarrow(n)}$, satisfy:

$$X_u^{\downarrow(n)} \leq X_u^{\downarrow(n+1)} \leq X_u \leq X_u^{\uparrow(n+1)} \leq X_u^{\uparrow(n)},$$

for all $u \in [0, 1]$, and $\sup_u |X_u^{\downarrow(n)} - X_u^{\uparrow(n)}| \rightarrow 0$ almost surely as $n \rightarrow \infty$.

The motivation behind this method is to enable unbiased Monte Carlo estimation for functionals depending on the entire Brownian path, such as extrema, hitting times, or integrals. Standard methods, like the Euler–Maruyama scheme, approximate the path at a finite grid of points, which introduces discretization bias. In contrast, the ε -strong algorithm allows for exact simulation by refining the bounding intervals until a stopping criterion is met, thereby avoiding such bias entirely.

This algorithm could also be extended to simulate paths of more general stochastic differential equations.

4.1 Intersection Layers

Intersection layers are the core structure of the algorithm, since they contain information about minima and maxima of the Brownian path in a subinterval. We define them and show how they are used in the algorithm.

Given a Brownian bridge X_t , let

$$m_{s,t} := \inf\{X_u; u \in [s, t]\}, \quad M_{s,t} := \sup\{X_u; u \in [s, t]\}.$$

Let also $\mathcal{U}_{s,t} = [U_{s,t}^{\uparrow}, U_{s,t}^{\downarrow}]$ and $\mathcal{L}_{s,t} = [L_{s,t}^{\uparrow}, L_{s,t}^{\downarrow}]$ such that $M_{s,t} \in \mathcal{U}_{s,t}$ and $m_{s,t} \in \mathcal{L}_{s,t}$. With these, we can define the intersection layer.

Definition 4.1. An *intersection layer* $\mathcal{I}_{s,t}$ is a tuple containing information on the Brownian bridge. In particular,

$$\mathcal{I}_{s,t} = \{X_s, X_t, \mathcal{L}_{s,t}, \mathcal{U}_{s,t}\}.$$

To run the algorithm, we need to be able to adjust the width of the layers $\mathcal{L}_{s,t}, \mathcal{U}_{s,t}$ relatively to $t - s$, to ensure convergence. We call this operation *refinement*. We present it for m , but the algorithm for refining M is analogous.

Algorithm 1 Refine($\mathcal{I}_{s,t}, m$)

- 1: Let $L^\downarrow$ and $L^\uparrow$ be the current lower and upper bounds on the minimum m over $[s, t]$. Let $L_{\text{mid}} = (L^\downarrow + L^\uparrow)/2$
 - 2: Sample a binary random variable to determine whether $m \in [L^\downarrow, L_{\text{mid}}]$ or $[L_{\text{mid}}, L^\uparrow]$
 - 3: **if** $m \in [L^\downarrow, L_{\text{mid}}]$ **then**
 - 4: Update $L^\uparrow \leftarrow L_{\text{mid}}$
 - 5: **else**
 - 6: Update $L^\downarrow \leftarrow L_{\text{mid}}$
 - 7: **end if**
 - 8: Update the layer $\mathcal{L}_{s,t} \leftarrow [L^\downarrow, L^\uparrow]$
 - 9: **return** updated $\mathcal{I}_{s,t}$
-

With the following algorithm, we granularize $\mathcal{I}_{s,t}$ into a more detailed information $\mathcal{I}_{s,t^*} \vee \mathcal{I}_{t^*,t}$ for an intermediate (i.e. middle) time t^* . We will get on how to sample X_{t^*} in Sections 4.3 and 4.4

Algorithm 2 Bisect($\mathcal{I}_{s,t}$)

- 1: Set $t^* = (t + s)/2$. Simulate X_{t^*} . Given $\mathcal{I}_{s,t}$, set $U^\downarrow = U^\downarrow \vee X_{t^*}$, $L^\uparrow = L^\uparrow \wedge X_{t^*}$
 - 2: Decide $\mathcal{U}_{s,t^*} [X_s \vee X_{t^*}, U^\downarrow]$ or $[U^\downarrow, U^\uparrow]$.
 - 3: Decide $\mathcal{L}_{s,t^*} = [L^\downarrow, L^\uparrow]$ or $[L^\uparrow, X_s \wedge X_{t^*}]$.
 - 4: Decide $\mathcal{U}_{t^*,t} = [X_{t^*} \vee X_t, U^\downarrow]$ or $[U^\downarrow, U^\uparrow]$.
 - 5: Decide $\mathcal{L}_{t^*,t} = [L^\downarrow, L^\uparrow]$ or $[L^\uparrow, X_{t^*} \wedge X_t]$.
 - 6: **return** $\mathcal{I}_{s,t^*} \vee \mathcal{I}_{t^*,t}$.
-

4.2 ε -strong Simulation

In this section, we present the main iterative simulation algorithm with input a Brownian bridge X on the domain $[0, 1]$ and which output, after n iterations, upper and lower dominating processes $X^\downarrow(n) = \{X_u^\downarrow(n); u \in [0, 1]\}$ and $X^\uparrow(n) = \{X_u^\uparrow(n); u \in [0, 1]\}$ satisfying the monotonicity and limiting requirements.

From the ε -strong algorithm, we can define the bounding processes as follows:

$$X_u^\uparrow(n) = \sum_{i=1}^{2^n} U_{(i-1)2^{-n}, i2^{-n}}^\uparrow \cdot \mathbb{1}_{u \in ((i-1)2^{-n}, i2^{-n}]} ; \quad X_u^\downarrow(n) = \sum_{i=1}^{2^n} L_{(i-1)2^{-n}, i2^{-n}}^\downarrow \cdot \mathbb{1}_{u \in ((i-1)2^{-n}, i2^{-n}]}.$$

Algorithm 3 ε -strong(X_0, X_1, n)

```
1: Initialize  $\mathcal{U}_{0,1}, \mathcal{L}_{0,1}$ , set  $\mathcal{I}_{0,1} = \{X_0, X_1, \mathcal{U}_{0,1}, \mathcal{L}_{0,1}\}$ . Set  $\mathcal{P} = \{\mathcal{I}_{0,1}\}$  and  $i = 1$ .
2: while  $i \leq n$  do
3:   for all intersection layers  $\mathcal{I}_{s,t}$  in  $\mathcal{P}$  do
4:     Set  $t^* = (t + s)/2$ .
5:     Bisect  $\mathcal{I}_{s,t}$  into  $\mathcal{I}_{s,t^*}$  and  $\mathcal{I}_{t^*,t}$ .
6:     Refine  $\mathcal{I}_{s,t^*}$  and  $\mathcal{I}_{t^*,t}$  until the width of their layers is not greater than  $\sqrt{(t - s)/2}$ .
7:   end for
8:   Collect the updated information:  $\mathcal{P} = \bigcup_{j=1}^{2^i} \mathcal{I}_{(j-1)2^{-i}, j2^{-i}}$ .
9:   Set  $i = i + 1$ .
10: end while
11: return  $\mathcal{P}$ 
```

4.2.1 Convergence Properties

Proposition 4.2. Consider the continuous-time processes $X^\uparrow(n), X^\downarrow(n)$ defined above. Then, the convergence in the supremum norm holds:

$$\lim_{n \rightarrow \infty} \sup_{u \in [0,1]} |X_u^\uparrow(n) - X_u^\downarrow(n)| = 0 \quad \text{a.s.}$$

Proposition 4.3. Consider the L_1 -distance:

$$|X^\uparrow(n) - X^\downarrow(n)|_1 = \int_0^1 |X_u^\uparrow(n) - X_u^\downarrow(n)| du$$

Then:

$$2^{n/2} E(|X^\uparrow(n) - X^\downarrow(n)|_1) = \mathcal{O}(1).$$

Proofs of these propositions can be found on [BPR11].

4.3 The ζ -function and Probabilities

We have yet to present the sampling methods employed when refining or bisecting an intersection layer. All probabilities involved in these methods can be expressed in terms of the hitting probability of the Brownian path.

4.3.1 ζ -function

Let $\mathbb{W}^{(l,x,y)}$ denote the probability law of a Brownian bridge from $X_0 = x$ to $X_l = y$.

Let $\zeta(L, U; l, x, y)$, with $L < U$, be the probability that the Brownian bridge escapes the interval $[L, U]$.

Thus:

$$\zeta(L, U; l, x, y) = \mathbb{W}^{(l,x,y)}[m_{0,l} < L \text{ or } M_{0,l} > U].$$

Define also $\gamma(L, U; l, x, y) := 1 - \zeta(L, U; l, x, y)$.

These probabilities can be calculated analytically in terms of an infinite series. In particular, we define for $j \geq 1$,

$$\begin{aligned}\bar{\sigma}_j(x, y, \delta, \xi) &= \exp \left\{ -\frac{2}{l} [\delta j + \xi - x] [\delta j + \xi - y] \right\}; \\ \bar{\tau}_j(x, y, \delta) &= \exp \left\{ -\frac{2j}{l} [\delta^2 j + \delta(x - y)] \right\}.\end{aligned}$$

Then, the paper concludes

$$\zeta(L, U; l, x, y) = \begin{cases} \sum_{j=1}^{\infty} (\sigma_j - \tau_j), & \text{if } L < x, y < U, \\ 1 & \text{otherwise,} \end{cases}$$

where

$$\begin{aligned}\sigma_j &= \bar{\sigma}_j(x, y, U - L, L) + \bar{\sigma}_j(-x, -y, U - L, -U), \\ \tau_j &= \bar{\tau}_j(x, y, U - L) + \bar{\tau}_j(-x, -y, U - L).\end{aligned}$$

The infinite series above exhibits a monotonicity property that will be exploited by our simulation algorithms. We consider the sequence $\{S_n\}$, with $S_n = S_n(L, U; l, x, y)$, defined as:

$$S_{2n-1} = \sum_{j=1}^{n-1} (\sigma_j - \tau_j) + \sigma_n, \quad S_{2n} = S_{2n-1} - \tau_n,$$

when $L < x, y < U$, otherwise $S_n := 1$. Then, for $n \geq 1$:

$$0 < S_{2n} \leq S_{2n+2} \leq \zeta \leq S_{2n+1} \leq S_{2n-1}. \quad (4.1)$$

4.3.2 ζ -Derived events

Now we can combine ζ -probabilities to calculate conditional probabilities arising in the context of the ε -strong algorithm.

First, we define

$$\beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; l, x, y) := \mathbb{W}^{(l, x, y)}[L^\downarrow < m_{0,l} < L^\uparrow, U^\downarrow < M_{0,l} < U^\uparrow].$$

Then, note the set equality

$$\begin{aligned}\{L^\downarrow < m_{0,l} < L^\uparrow, U^\downarrow < M_{0,l} < U^\uparrow\} \\ = \{L^\downarrow < m_{0,l}, M_{0,l} < U^\uparrow\} - \{L^\uparrow < m_{0,l}, M_{0,l} < U^\uparrow\} \cup \{L^\downarrow < m_{0,l}, M_{0,l} < U^\downarrow\}.\end{aligned}$$

Taking probabilities and recalling the definition of γ , we get

$$\beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; l, x, y) = \gamma(L^\downarrow, U^\uparrow; l, x, y) - \gamma(L^\uparrow, U^\uparrow; l, x, y) - \gamma(L^\downarrow, U^\downarrow; l, x, y) + \gamma(L^\uparrow, U^\downarrow; l, x, y),$$

which is the probability that the Brownian bridge does not escape the interval $[L, U]$.

For the next event, we first define

$$\mathbb{W}_{(q,w)}^{(l,x,y)}[\cdot] := \mathbb{W}^{(l,x,y)}[\cdot | X_q = w] \quad \text{for } 0 < q < l.$$

Set $r := l - q$ and consider

$$\rho(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, r, x, w, y) = \mathbb{W}_{(q,w)}^{(l,x,y)}[L^\downarrow < m_{0,l} < L^\uparrow, U^\downarrow < M_{0,l} < U^\uparrow].$$

Using the same set equality and taking probabilities, we get

$$\begin{aligned} \rho(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, r, x, w, y) = & \\ & \gamma(L^\downarrow, U^\uparrow; q, x, w) \gamma(L^\downarrow, U^\uparrow; r, w, y) - \gamma(L^\uparrow, U^\uparrow; q, x, w) \gamma(L^\uparrow, U^\uparrow; r, w, y) \\ & - \gamma(L^\downarrow, U^\downarrow; q, x, w) \gamma(L^\downarrow, U^\downarrow; r, w, y) + \gamma(L^\uparrow, U^\downarrow; q, x, w) \gamma(L^\uparrow, U^\downarrow; r, w, y). \end{aligned} \quad (4.2)$$

Product terms arise due to the independence of the Brownian bridges on $[0, q]$ and $[q, l]$. In the following section, we will use these expressions for $\beta(\cdot; \cdot)$ and $\rho(\cdot; \cdot)$.

4.3.3 Simulation of ζ -derived events

We want to be able to decide whether events of probability ζ have occurred or not. In a simulation context, this corresponds to determining the value of the binary variable $\mathbb{1}_{R < \zeta}$ for $R \sim \text{Unif}[0, 1]$. Define

$$J := \inf\{n \geq 1 : \{n \text{ odd}, S_n < R\} \vee \{n \text{ even}, S_n > R\}\}.$$

Thanks to (4.1) (alternating monotonicity property of S_n), $\mathbb{1}_{R < \zeta} = \mathbb{1}_{J \text{ is even}}$. Thus, we need a.s. a finite number of J computations to evaluate $\mathbb{1}_{R < \zeta}$. Note also that S_n converges to its limit exponentially fast, so J will be of small expectation; one can easily verify that all its moments are finite.

In a more general context, we will also be required to decide if events of probability $\beta(\cdot; \cdot)$ or $\rho(\cdot; \cdot)$ have taken place.

Using the monotonicity property, we will be able to develop corresponding alternating sequences S_n^Z such that:

$$S_{2n}^Z \leq S_{2n+2}^Z \leq Z(\zeta_1, \zeta_2, \dots, \zeta_m) \leq S_{2n+1}^Z \leq S_{2n-1}^Z; \quad \lim_{n \rightarrow \infty} S_n^Z = Z(\zeta_1, \zeta_2, \dots, \zeta_m), \quad (4.3)$$

for some given function Z .

Practically, we will calculate S_n^Z until the first n such that either n is odd and $S_n^Z < R$ (returning 0) or n is even and $S_n^Z > R$ (returning 1).

4.4 Simulation of Distributions

We now describe in detail all the simulation algorithms employed. In our case, we can use variables as follows:

$$x = X_s, \quad w = X_{t^*}, \quad y = X_t, \quad l = t - s, \quad q = t^* - s, \quad r = t - t^*.$$

4.4.1 Bisection: Sampling X_{t^*}

The first step of the bisection algorithm requires sampling a point from a Brownian bridge conditionally on some information on the minimum and maximum. This can be computed analytically:

Proposition 4.4. The distribution $\mathbb{W}[X_{t^*} | \mathcal{I}_{s,t}]$, with $t^* \in [s, t]$ has density

$$f(w) \propto \rho(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, r, x, w, y) \cdot \pi(w),$$

with $\rho(\cdot; \cdot)$ defined as in (4.2) and

$$\pi(w) = \exp\left(-\frac{1}{2} \left(w - \left(\frac{r}{l}x + \frac{q}{l}y\right)\right)^2 / \left(\frac{qr}{l}\right)\right).$$

For a proof, see [BPR11], Section 5.

To sample from f , we construct an alternating series bounding it. Let

$$\zeta_i = 1 - \gamma_i, \quad 1 \leq i \leq 8,$$

with γ_i being the i -th term of the product in (4.2). Let S be an alternating sequence as in (4.1). Then,

$$0 \leq S_{i,2n} \leq S_{i,2n+2} \leq \zeta_i \leq S_{i,2n+1} \leq S_{i,2n-1}, \quad \lim_{n \rightarrow \infty} S_{i,n} = \zeta_i.$$

If we consider the sequence $\{S_n^Z\}$ defined as

$$\begin{aligned} S_n^Z &= (1 - S_{1,n+1} - S_{2,n+1} + S_{1,n}S_{2,n}) - (1 - S_{3,n} - S_{4,n} + S_{3,n+1}S_{4,n+1}) \\ &\quad - (1 - S_{5,n} - S_{6,n} + S_{5,n+1}S_{6,n+1}) - (1 - S_{7,n+1} - S_{8,n+1} + S_{7,n}S_{8,n}), \end{aligned}$$

one can verify that this is an alternating sequence for $\rho(\cdot; \cdot)$, i.e.:

$$S_{2n}^Z \leq S_{2n+2}^Z \leq \rho(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, r, x, w, y) \leq S_{2n+1}^Z \leq S_{2n-1}^Z, \quad \lim_{n \rightarrow \infty} S_n^Z = \rho(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, r, x, w, y).$$

Now we can exploit $\{S_n^Z\}$ to build a rejection sampler drawing from the density of $f(w)$. We use $f_{2n+1}(w) = S_{2n+1}^Z(w) \cdot \pi(w)$.

The paper shows that f_{2n+1} can be rewritten as

$$f_{2n+1}(w) = \sum_{i=1}^{k_n} (-1)^{c_i} \exp\{a_i + b_i w\} \mathbb{1}_{L_i < w < U_i} \cdot \pi(w)$$

Empirically, f_1 proved to be a good function for the rejection sampler. Thus, we set $k_n := k_0 = 64$. This is due to the exponential convergence of the alternating sequence in the limit.

Turns out we can generate samples for $f_1(w)$ using the inverse cdf method, i.e., returning $F_1^{-1}(R)$ for $R \sim \text{Unif}[0, 1]$. We can approximate it using numerical methods.

Hence, the sampler will look like:

Repeat until the first accepted draw:

Propose $w \sim f_1$ and accept with probability $f(w)/f_1(w) := \rho(w)/S_1^Z(w)$.

4.4.2 Bisection: Updating Layers given X_{t^*}

At the second step of the bisection, we obtain separate information for the extrema of the two Brownian bridges given the middle point X_{t^*} . The algorithm will decide over the range of four new layers: $\mathcal{L}_{s,t^*}, \mathcal{L}_{t^*,t}, \mathcal{U}_{s,t^*}, \mathcal{U}_{t^*,t}$. For instance, in the case of L_{s,t^*} , a decision will be made over whether m_{s,t^*} lies in $[L^\downarrow, L^\uparrow]$, which is the allowed range for the minimum of the original bridge on $[s, t]$, or in $[L^\uparrow, X_s \wedge X_{t^*}]$. The procedure is analogous for the other layers. In principle, there are 2^4 possible scenarios for the layers, but since they must respect information contained in $\mathcal{I}_{s,t}$, we end up having nine possible scenarios, illustrated in Table 4.1, Where '1' means that the corresponding minimum or maximum will still be found within the allowed range for the original bridge, whereas 0 means that the second option occurs and the extremum will be shifted inwards.

Table 4.1: The nine possible scenarios for the extrema of the two Brownian bridges

Event $E = i$	Left bridge		Right bridge	
	$\mathbb{1}_{m_{s,t^*} \in [L^\downarrow, L^\uparrow]}$	$\mathbb{1}_{M_{s,t^*} \in [U^\downarrow, U^\uparrow]}$	$\mathbb{1}_{m_{t^*,t} \in [L^\downarrow, L^\uparrow]}$	$\mathbb{1}_{M_{t^*,t} \in [U^\downarrow, U^\uparrow]}$
$i = 1$	1	1	1	1
$i = 2$	1	1	0	1
$i = 3$	1	1	1	0
$i = 4$	1	1	0	0
$i = 5$	0	1	1	1
$i = 6$	0	1	1	0
$i = 7$	1	0	1	1
$i = 8$	1	0	0	1
$i = 9$	0	0	1	1

The probability for each of the events can be derived via functions $\beta(\cdot; \cdot)$ and $\rho(\cdot; \cdot)$:

$$\begin{aligned} P[E = i \mid \mathcal{I}_{s,t}, X_{t^*}] &= P[E = i \mid m \in [L^\downarrow, L^\uparrow], M \in [U^\downarrow, U^\uparrow], X_s, X_{t^*}, X_t] \\ &= \frac{P[E = i, m \in [L^\downarrow, L^\uparrow], M \in [U^\downarrow, U^\uparrow] \mid X_s, X_{t^*}, X_t]}{P[m \in [L^\downarrow, L^\uparrow], M \in [U^\downarrow, U^\uparrow] \mid X_s, X_{t^*}, X_t]} \\ &= \frac{P[E = i \mid X_s, X_{t^*}, X_t]}{\rho(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, r, x, w, y)}. \end{aligned}$$

Conditioning on X_s, X_{t^*}, X_t , the law of the path factorizes by independence of the two bridges. Thus, we can extract probabilities for $P[E = i \mid X_s, X_{t^*}, X_t]$ in Table 4.2.

Table 4.2: The conditional probabilities for each of the events in Table 4.1

i	$P[E = i \mid \mathcal{I}_{s,t}, X_{t^*}] \cdot \rho(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, r, x, w, y)$
1	$\beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, x, y) \cdot \beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; r, w, y)$
2	$\beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, x, w) \cdot \beta(L^\uparrow, y \wedge w, U^\downarrow, U^\uparrow; r, w, y)$
3	$\beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, x, w) \cdot \beta(L^\downarrow, L^\uparrow, y \vee w, U^\downarrow; r, w, y)$
4	$\beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, x, w) \cdot \beta(L^\uparrow, y \wedge w, y \vee w, U^\downarrow; r, w, y)$
5	$\beta(L^\uparrow, x \wedge w, U^\downarrow, U^\uparrow; q, x, w) \cdot \beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; r, w, y)$
6	$\beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; q, x, w) \cdot \beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; r, w, y)$
7	$\beta(L^\uparrow, x \wedge w, U^\downarrow, U^\uparrow; q, x, w) \cdot \beta(L^\downarrow, L^\uparrow, y \vee w, U^\downarrow; r, w, y)$
8	$\beta(L^\downarrow, L^\uparrow, x \vee w, U^\downarrow; q, x, w) \cdot \beta(L^\uparrow, y \wedge w, U^\downarrow, U^\uparrow; r, w, y)$
9	$\beta(L^\uparrow, x \wedge w, x \vee w, U^\downarrow; q, x, w) \cdot \beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; r, w, y)$

The method for simulating the discrete random variable E uses inverse transform sampling, where one finds the smallest index i such that a uniform random variable $R \sim \text{Unif}[0, 1]$ satisfies $R < p_i$, with $p_i = P[E \leq i \mid \mathcal{I}_{s,t}, X_{t^*}]$. Since the cumulative probabilities p_i can be bounded above and below by monotone sequences that converge (as in equation (4.3)), each comparison $\{R < p_i\}$ can be performed using the alternating series approach from Section 4.3.3, without requiring any numerical approximation.

4.4.3 Refinement

A sampling algorithm is required for the refinement of the uncertainty over the extrema of a Brownian bridge. Given $\mathcal{I}_{s,t}$ and $M_{s,t} \in [L^\downarrow, L^\uparrow]$, the algorithm should decide whether $M_{s,t} \in [U^\downarrow, U^*]$ or $M_{s,t} \in [U^*, U^\uparrow]$, with $U^* = \frac{U^\downarrow + U^\uparrow}{2}$.

The ratio

$$\frac{\beta(L^\downarrow, L^\uparrow, U^*, U^\uparrow; l, x, y)}{\beta(L^\downarrow, L^\uparrow, U^\downarrow, U^\uparrow; l, x, y)}$$

corresponds to $P[M_{s,t} \in [U^*, U^\uparrow] \mid \mathcal{I}_{s,t}]$. Thus, we can use the same alternating sequence construction as Section 4.3.3 to simulate, without approximation, the binary variable $\mathbb{1}_{M_{s,t} \in [U^*, U^\uparrow]}$. The same goes for $m_{s,t}$.

Appendix A

Measure-Theoretic Probability

In this section, we introduce fundamental concepts and construct the best framework to study stochastic processes.

A.1 Probability Spaces and Lebesgue-Stieltjes Measure

We proceed by defining a series of objects necessary for the development of the framework. Random variables live in a space, called a probability space, and they enjoy some properties. In this section, we introduce them.

A.1.1 Measurable spaces

The first space we consider is the *sample space* Ω , which is the set of all possible outcomes of an experiment.

Definition A.1. A collection of subsets $\mathcal{F}$ of Ω is called an *algebra* or a *field* if:

1. $\Omega \in \mathcal{F}$,
2. $A \in \mathcal{F}, B \in \mathcal{F} \implies A \cup B \in \mathcal{F}$,
3. $A \in \mathcal{F} \implies A^c \in \mathcal{F}$.

A in $\mathcal{F}$ is called an *event*.

Using de Morgan's law, we can also deduce that an algebra is closed under (finitely many) intersections and unions.

In probability theory, computing the probability that a set of events occur simultaneously is allowed as long as the events involved are at most countable.

Definition A.2. A collection of subsets $\mathcal{F}$ of Ω is called a σ -*algebra* or a σ -*field* if:

1. $\Omega \in \mathcal{F}$,
2. Given $(A_i)_{i=1}^{\infty}$, if $\forall i, A_i \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$,
3. $A \in \mathcal{F} \implies A^c \in \mathcal{F}$.

An element in $\mathcal{F}$ is called $\mathcal{F}$ -*measurable* (sometimes we refer to it simply as *measurable*).

$(\Omega, \mathcal{F})$ is a *measurable space* if $\mathcal{F}$ is a σ -algebra of Ω .

Definitions A.1 and A.2 coincide if Ω is finite. In the infinite case, an algebra might not be a σ -algebra.

A.1.2 Measure functions

Now that we know what a σ -algebra $\mathcal{F}$ is, we can define measure functions, which take $\mathcal{F}$ as domain.

Definition A.3. Let $\mathcal{F}$ be a σ -algebra on a sample space Ω . A set function $m : \mathcal{F} \rightarrow [0, \infty]$ is called *measure* if:

1. $m(\emptyset) = 0$,
2. σ -additivity: $\forall (A_i)_{i=1}^{\infty}$ of mutually disjoint $A_i \in \mathcal{F}$, we have $m(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} m(A_i)$

A measure m is said to be *finite* if $m(\Omega)$ is finite. If $m(\Omega) = 1$, then m is called a *probability measure*. A measure space is a triple $(\Omega, \mathcal{F}, m)$ where $\mathcal{F}$ is a σ -algebra on Ω and m is a measure on $\mathcal{F}$. A *probability space* is a measure space $(\Omega, \mathcal{F}, m)$, with m a probability measure.

Theorem A.4. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Suppose A, B are $\mathcal{F}$ -measurable sets. Then, the measure μ has the following properties:

1. *Monotonicity:* if $A \subseteq B$. Then, $\mu(A) \leq \mu(B)$,
2. *Finite Subadditivity:* $\mu(A \cup B) \leq \mu(A) + \mu(B)$,
3. *Lower Semi-Continuity:* Let $A_1 \subseteq A_2 \subseteq \dots$ be a sequence in $\mathcal{F}$. Then $\lim_{k \rightarrow \infty} \mu(A_k) = \mu(\bigcup_{i=1}^{\infty} A_i)$,
4. *Upper Semi-Continuity:* Let $A_1 \supseteq A_2 \supseteq \dots$ be a sequence in $\mathcal{F}$ and $\mu(A_1) < \infty$. Then $\lim_{k \rightarrow \infty} \mu(A_k) = \mu(\bigcap_{i=1}^{\infty} A_i)$.

Proof. They all follow from simple set properties. For a reference, see [Shu23] section 2.3. $\square$

A.1.3 Borel Sets

In a general topological space, it is not always easy to define a σ -algebra. Indeed, the open sets in the topological space may not satisfy all the requirements of a σ -algebra. In this case, we define the *Borel algebra* as the smallest σ -algebra that contains all open sets in the given topological space.

Theorem A.5. Let $\mathcal{C}$ be a collection of subsets in Ω , and $\mathcal{F}$ be the intersection of all σ -algebras that contain $\mathcal{C}$ as a sub-collection. Then $\mathcal{F}$ is a σ -algebra, and it is the smallest σ -algebra that contains $\mathcal{C}$.

Proof. We can write

$$\mathcal{F} = \bigcap_{\mathcal{G}: \sigma\text{-algebra}; \mathcal{C} \subseteq \mathcal{G}} \mathcal{G}$$

We define $\mathcal{F}$ as the intersection of all σ -algebras containing $\mathcal{C}$. This intersection is non-empty since $\mathcal{P}(\Omega)$ (the power set) is a σ -algebra. We verify that $\mathcal{F}$ satisfies the three axioms:

- Since Ω belongs to every σ -algebra, we have $\Omega \in \mathcal{F}$.

- If $A \in \mathcal{F}$, then A is in every σ -algebra containing $\mathcal{C}$. As σ -algebras are closed under complements, $A^c \in \mathcal{F}$.
- If $A_i \in \mathcal{F}$ for $i = 1, 2, \dots$, then A_i belongs to every σ -algebra containing $\mathcal{C}$, so $\bigcup_i A_i \in \mathcal{F}$.

Thus, $\mathcal{F}$ is a σ -algebra containing $\mathcal{C}$. If $\mathcal{G}$ is any σ -algebra containing $\mathcal{C}$, then $\mathcal{F} \subseteq \mathcal{G}$, proving that $\mathcal{F}$ is the smallest σ -algebra containing $\mathcal{C}$. $\square$

Such $\mathcal{F}$ is called the σ -algebra generated by $\mathcal{C}$. We denote it $\sigma(\mathcal{C})$.

Definition A.6. Let $\Omega = \mathbb{R}$ and $\mathcal{C}$ be the collection of all open intervals (a, b) with $a < b$. The σ -algebra generated by $\mathcal{C}$ is called the *Borel σ -field* or the *Borel σ -algebra*, and is denoted by $\mathcal{B}(\mathbb{R})$. Likewise, for integer $d \geq 1$, we define the *Borel σ -algebra* $\mathcal{B}(\mathbb{R}^d)$ as the σ -algebra generated by the open balls in $\mathbb{R}^d$. A set in a Borel σ -algebra is called a *Borel set*.

The Borel σ -algebra $\mathcal{B}(\mathbb{R})$ contains all intervals one needs in practice.

It has been proved that even with the simplest uniform distribution, there exists a set to which we cannot assign any probability. Instead, we assign probabilities only to the subsets in a σ -algebra, and subsets that are not in the σ -algebra have no probability.

A.1.4 Lebesgue–Stieltjes Measures

We now state an important theorem for extending pre-measures to measures in a σ -algebra. First, we need to define what a *pre-measure* is. It is a set function that takes a set as input and returns a nonnegative real number. It satisfies σ -additivity, but unlike a measure function, the domain of a pre-measure is an algebra or a field, rather than a σ -algebra.

Theorem A.7 (Hahn–Kolmogorov–Carathéodory).

Let $\mathcal{F}_0$ be a field on a sample space Ω . Given a pre-measure μ_0 defined on $\mathcal{F}_0$, there is a measure μ that extends to $\sigma(\mathcal{F}_0)$.

Moreover, if μ_0 is σ -finite, then the extension is unique, i.e., if μ and μ' are two extensions of μ_0 , we have $\mu(E) = \mu'(E) \ \forall E \in \mathcal{F}$.

Proof. For the existence, see [Dur19], Theorem 1.1.9. Uniqueness is proved in A.1.5 $\square$

Definition A.8. Let $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P)$ be a probability space. The *distribution function induced by P* is

$$F(x) := P((-\infty, x]), \quad x \in \mathbb{R}.$$

Theorem A.9. The distribution function $F(x)$ of a probability measure P satisfies:

- (i) $F(x)$ non-decreasing,

(ii) $F(x)$ right-continuous, i.e. $\lim_{y \rightarrow x^+} F(y) = F(x) \forall x \in \mathbb{R}$,

(iii) $\lim_{x \rightarrow \infty} F(x) = 1$,

(iv) $\lim_{x \rightarrow -\infty} F(x) = 0$.

Proof. (i) and (ii) follow from monotonicity and upper semi-continuity of a measure, respectively. If $(x_i)_{i \in \mathbb{N}}$ is a decreasing sequence converging to x from the right, then $(-\infty, x_i]$ is a decreasing sequence of sets, and the intersection is $(-\infty, x]$. Hence, $\lim_{i \rightarrow \infty} F(x_i) = \lim_{i \rightarrow \infty} P((-\infty, x_i]) = P((-\infty, x]) = F(x)$. Letting $x \rightarrow \pm\infty$ (and using upper/lower semi-continuity), one proves (iii) and (iv). $\square$

Now we have the tools to define the Lebesgue–Stieltjes measure. First, we define a measure on $\mathcal{B}(\mathbb{R})$.

Definition A.10. A function $F : \mathbb{R} \rightarrow \mathbb{R}$ is called *Stieltjes measure function* if:

(i) F is non-decreasing,

(ii) F is right-continuous.

In general, a Stieltjes measure function defines a measure on the Borel algebra of $\mathbb{R}$, which is called the Lebesgue–Stieltjes measure.

Theorem A.11. Let F be a Stieltjes measure function defined on $\mathbb{R}$. Then there exists a unique measure μ defined on Borel σ -algebra $\mathcal{B}(\mathbb{R})$ such that $\mu((a, b]) = F(b) - F(a)$. The measure μ is called the *Lebesgue–Stieltjes measure induced by F* .

Proof. It relies on the previous extension theorem A.7 and Heine-Borel theorem. See [Shu23], Theorem 3.3. $\square$

A.1.5 Uniqueness of measure extension

Definition A.12. We define a π -system in Ω as a collection of subsets (of Ω) $\mathcal{P}$ that is closed under intersection, i.e., it satisfies

$$A \in \mathcal{P} \text{ and } B \in \mathcal{P} \Rightarrow A \cap B \in \mathcal{P}.$$

A λ -system in Ω is a collection of subsets $\mathcal{L}$ that satisfies the following conditions:

(i) $\Omega \in \mathcal{L}$.

(ii) If $A \in \mathcal{L}$, then $\Omega \setminus A \in \mathcal{L}$.

(iii) If $A_i \in \mathcal{L}$ for $i = 1, 2, 3, \dots$ is a sequence of mutually disjoint sets, then $\bigcup_i A_i \in \mathcal{L}$.

A λ -system is also known as a Dynkin system, named after Eugene Dynkin, who first introduced this concept. The π - λ theorem is a fundamental result in measure theory and probability theory, which will be crucial to prove the uniqueness of measure extension. For the sake of brevity, we do not provide a formal proof of the theorem.

Theorem A.13. *If $\mathcal{L}$ is a λ -system containing a π -system $\mathcal{P}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$.*

We can now use the π - λ theorem to prove the uniqueness of measure extension. The theorem states that if two measures on $\sigma(\mathcal{F}_0)$ agree on a field $\mathcal{F}_0$ that generates $\sigma(\mathcal{F}_0)$ and also agree on a sequence of disjoint sets that cover the entire sample space, then they agree on $\sigma(\mathcal{F}_0)$.

Theorem A.14 (Uniqueness of Measure Extension).

Suppose $\mathcal{F}_0$ is an algebra on a sample space Ω , and μ_1 and μ_2 are two measures on $\sigma(\mathcal{F}_0)$ such that $\mu_1(A) = \mu_2(A)$ for all $A \in \mathcal{F}_0$. Furthermore, suppose there exists a sequence of disjoint sets $\Omega_i \in \mathcal{F}_0$ for $i = 1, 2, 3, \dots$ such that

$$\bigcup_i \Omega_i = \Omega \quad \text{and} \quad \mu_1(\Omega_i) = \mu_2(\Omega_i) < \infty \quad \forall i.$$

Then $\mu_1(B) = \mu_2(B)$ for all $B \in \sigma(\mathcal{F}_0)$.

Proof. The theorem states that if two measures on $\sigma(\mathcal{F}_0)$ agree on an algebra $\mathcal{F}_0$ that generates $\mathcal{F}$ and also agree on a sequence of disjoint sets that cover Ω , then they agree on $\mathcal{F}$.

Let S be a set in $\mathcal{F}_0$ such that $\mu_1(S) = \mu_2(S) < \infty$. Consider the collection of subsets

$$\mathcal{M}_S = \{Q \in \sigma(\mathcal{F}_0) : \mu_1(S \cap Q) = \mu_2(S \cap Q)\}.$$

We claim that

- (a) $\mathcal{F}_0 \subseteq \mathcal{M}_S$;
- (b) $\mathcal{M}_S$ is a λ -system.

We first show that the measure extension is unique given the claims in (a) and (b). Then, we will prove (a) and (b). Since $\mathcal{F}_0$ is certainly a π -system, by the π - λ theorem, we have

$$\mathcal{M}_S \supseteq \sigma(\mathcal{F}_0).$$

We apply it with S replaced by Ω_i , for $i = 1, 2, 3, \dots$. This yields $\sigma(\mathcal{F}_0) \subseteq \mathcal{M}_{\Omega_i}$ for all i . Let E be any event in $\sigma(\mathcal{F}_0)$. We can write E as

$$E = E \cap \Omega = E \cap \left(\bigcup_{i=1}^{\infty} \Omega_i \right) = \bigcup_{i=1}^{\infty} (E \cap \Omega_i).$$

Since μ_1 and μ_2 are measure functions, they are σ -additive. Hence,

$$\mu_k(E) = \sum_{i=1}^{\infty} \mu_k(E \cap \Omega_i)$$

for $k = 1, 2$. Since $E \in \sigma(\mathcal{F}_0) \subseteq \mathcal{M}_{\Omega_i}$ for all i , we have

$$\mu_1(E \cap \Omega_i) = \mu_2(E \cap \Omega_i)$$

for all i , by the definition of $\mathcal{M}_{\Omega_i}$. This implies that $\mu_1(E) = \mu_2(E)$.

To prove (a), suppose $A \in \mathcal{F}_0$. Then $A \cap S \in \mathcal{F}_0$ because both A and S are in $\mathcal{F}_0$ and $\mathcal{F}_0$ is closed under intersection. By the assumption on μ_1 and μ_2 , we have $\mu_1(A \cap S) = \mu_2(A \cap S)$. Therefore, $A \in \mathcal{M}_S$ for all $A \in \mathcal{F}_0$. This proves that $\mathcal{F}_0 \subseteq \mathcal{M}_S$.

To prove (b), we divide the argument into three parts.

(i) We have $\Omega \in \mathcal{M}_S$ because $\mu_1(\Omega \cap S) = \mu_1(S) = \mu_2(S) = \mu_2(\Omega \cap S)$.

(ii) Suppose $A \in \mathcal{M}_S$. Because S can be written as $S = (A \cap S) \cup (A^c \cap S)$, we have $\mu_1(S) = \mu_1(A \cap S) + \mu_1(A^c \cap S)$ for $k = 1, 2$. Using the assumption that $\mu_1(S) = \mu_2(S) < \infty$, we obtain

$$\mu_1(A^c \cap S) = \mu_1(S) - \mu_1(A \cap S) = \mu_2(S) - \mu_2(A \cap S) = \mu_2(A^c \cap S).$$

Therefore, $A^c \in \mathcal{M}_S$.

(iii) Suppose A_i are mutually disjoint sets in $\mathcal{M}_S$. This means that $\mu_1(A_i \cap S) = \mu_2(A_i \cap S)$ for all i . By σ -additivity, we obtain

$$\mu_1\left(\left(\bigcup A_i\right) \cap S\right) = \mu_2\left(\left(\bigcup A_i\right) \cap S\right) \Rightarrow \mu_1\left(\bigcup (A_i \cap S)\right) = \mu_2\left(\bigcup (A_i \cap S)\right).$$

This shows that $\bigcup A_i$ is in $\mathcal{M}_S$, completing the proof that $\mathcal{M}_S$ is a λ -system. $\square$

From this result, we can prove that a probability measure on $\mathbb{R}$ is uniquely determined by its Stieltjes measure function.

Theorem A.15. *Let P be a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, and let $F(x)$ be its induced distribution function. If there is another probability Q on $\mathcal{B}(\mathbb{R})$ such that*

$$Q((-\infty, x]) = P((-\infty, x]) \quad \forall x \in \mathbb{R},$$

then $P(B) = Q(B)$ for all $B \in \mathcal{B}(\mathbb{R})$.

Proof. We can apply the uniqueness of measure extension theorem to the algebra

$$\mathcal{B}_0 = \{\text{finite disjoint union of sets of the form } (a, b]\}.$$

When $a < b$, the probability of $(a, b]$ with respect to P is given by

$$P((a, b]) = P((-\infty, b]) - P((-\infty, a]) = F(b) - F(a).$$

By extending this to finite disjoint unions, we obtain a pre-measure, denoted by μ_0 , on the sets in $\mathcal{B}_0$. In fact, if $A = \bigcup_{i=1}^n (a_i, b_i]$, with $a_1 < b_1 \leq a_2 < b_2 \leq \dots \leq a_n < b_n$, then

$$\mu_0(A) = P(A) = \sum_{i=1}^n [F(b_i) - F(a_i)].$$

Suppose Q is a probability measure identical to P when restricted to sets in the form $(-\infty, a]$. Then for any $A \in \mathcal{B}_0$, we have

$$Q(A) = \sum_{i=1}^n [F(b_i) - F(a_i)] = P(A).$$

Since $\mathcal{B}(\mathbb{R})$ is the σ -algebra generated by $\mathcal{B}_0$, the extensions Q and P of μ_0 must be the same probability measure by A.14. $\square$

A.2 Random Variables

In probability theory, a random variable is a mathematical object that assigns numerical values to the outcomes of a random experiment. It serves as a bridge between abstract probability spaces and real-world quantities, allowing us to quantify uncertainty and make predictions.

In the following section, we formalize the concept of random variables as measurable functions, explore operations on them, and develop the necessary tools to analyze their behaviour.

A.2.1 Measurable functions

Definition A.16. let $(\Omega, \mathcal{F})$ be a measurable space. A function $X : \Omega \rightarrow \mathbb{R}$ is *measurable* if for every Borel set $B \in \mathcal{B}(\mathbb{R})$, we have

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$$

Note that the definition is independent of the probability measure chosen. If we introduce it and consider $(\Omega, \mathcal{F}, \mu)$, we call X a random variable.

With this definition, it is assured that we can assign probabilities to every set $\{\omega \in \Omega : X(\omega) \in B\}$. From now on we will denote the probability of such events as

$$P(\{\omega \in \Omega : X(\omega) \in B\}) = P(X(\omega) \in B) = P(X \in B)$$

In general, we can define a random variable that takes values in a set Ω' equipped with a σ -algebra $\mathcal{G}$.

Definition A.17. a function $f : (\Omega, \mathcal{F}) \rightarrow (\Omega', \mathcal{G})$ is called $(\mathcal{F}, \mathcal{G})$ -*measurable* if

$$f^{-1}(B) := \{\omega \in \Omega : f(\omega) \in B\}$$

is $\mathcal{F}$ -measurable for all $B \in \mathcal{G}$.

Example. If $\mathcal{F}$ is the trivial σ -algebra $\{\emptyset, \Omega\}$, and $\mathcal{G}$ is the σ -algebra in which all singletons are $\mathcal{G}$ -measurable, then an $(\mathcal{F}, \mathcal{G})$ -measurable function is a constant function. If $\mathcal{F}$ is the power set $\mathcal{P}(\Omega)$, then no matter what f is, all $(\mathcal{F}, \mathcal{G})$ functions from Ω to Ω' are measurable.

At this point, it remains to determine what σ -algebra we should select. Heuristically, one would want $\mathcal{F}$ as large as possible, to maximize flexibility, and $\mathcal{G}$ small. Often, when the codomain of the measurable function is $\mathbb{R}^n$, we usually take the $\mathcal{B}(\mathbb{R}^n)$ as σ -algebra in the codomain but do not necessarily consider a complete measure space in the codomain. By selecting different codomains, we can obtain various types of random variables.

Theorem A.18. *Let $(\Omega, \mathcal{F})$ be a measurable space and $A \subset \Omega$. Then the indicator function*

$$\mathbb{1}_A(\omega) := \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

is $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable $\iff A$ is $\mathcal{F}$ -measurable.

Proof. We consider four cases. Suppose B is a Borel set that contains 1 but not 0. Then the pre-image $\mathbb{1}_A^{-1}(B)$ is equal to the set A . If B is a Borel set that contains 0 but not 1, then $\mathbb{1}_A^{-1}(B) = A^c$. When B contains both 0 and 1, then $\mathbb{1}_A^{-1}(B) = \Omega$. Finally, when B does not contain 0 nor 1, then $\mathbb{1}_A^{-1}(B) = \emptyset$. This exhausts all possibilities of Borel sets B containing 0 or 1.

If A is $\mathcal{F}$ -measurable, then the four possible pre-images A , A^c , Ω , and $\emptyset$ are all in $\mathcal{F}$, and hence, the indicator function $\mathbb{1}_A$ is $\mathcal{F}$ -measurable. Otherwise, if A is not in $\mathcal{F}$, then the indicator function is not $\mathcal{F}$ -measurable. $\square$

A.2.2 Operations with measurable functions

First, we develop the theory for composition of measurable functions. We state, without proof, some properties of inverse image functions:

Proposition A.19. Suppose $f : \Omega \rightarrow \Omega'$ and A in any set in Ω' . Then:

- (1) $f^{-1}(A^c) = (f^{-1}(A))^c$;
- (2) $f^{-1}(\bigcap_{i \in I} A_i) = \bigcap_{i \in I} f^{-1}(A_i)$;
 $f^{-1}(\bigcup_{i \in I} A_i) = \bigcup_{i \in I} f^{-1}(A_i)$ with I possibly uncountably infinite;
- (3) If $h := g \circ f$, then $h^{-1}(B) = f^{-1}(g^{-1}(B))$ for any subset B of the codomain of g .

With these tools, we can give some important properties of measurable functions.

Theorem A.20 (Composition of measurable functions). *Suppose $f : (\Omega, \mathcal{F}) \rightarrow (\Omega', \mathcal{G})$ is $(\mathcal{F}, \mathcal{G})$ -measurable, and $g : (\Omega', \mathcal{G}) \rightarrow (\Omega'', \mathcal{H})$ is $(\mathcal{G}, \mathcal{H})$ -measurable. Then the composed function $h := g \circ f$ is $(\mathcal{F}, \mathcal{H})$ -measurable.*

Proof. Suppose A is a set in $\mathcal{H}$. Because g is $(\mathcal{G}, \mathcal{H})$ -measurable, then the pre-image $g^{-1}(A)$ is in $\mathcal{G}$. Since f is $(\mathcal{F}, \mathcal{G})$ -measurable, we have $f^{-1}(g^{-1}(A))$ is in $\mathcal{F}$. Hence, $h^{-1}(A) = f^{-1}(g^{-1}(A))$. $\square$

Remark. Some analysis textbooks state that the composition of two measurable functions may not be measurable. However, this statement refers to the Lebesgue σ -algebra, which is larger than the Borel σ -algebra used in our definition of measurability. The Lebesgue σ -algebra on $\mathbb{R}$, denoted $\mathcal{L}(\mathbb{R})$, is a complete measure constructed using the so-called *outer measure* and is much larger than the Borel algebra. Instead, the statement is false if the intermediate σ -algebra $(\Omega', \mathcal{G})$ is different for f and g . In that case, one must check that the σ -algebra on the “middle” space that f lands into is rich enough to make g measurable with respect to it. In other words,

$$(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B)) \in \mathcal{F}, \quad \forall B \in \mathcal{H}. \quad \text{or equivalently} \quad \mathcal{G}_g \subseteq \mathcal{G}_f,$$

Theorem A.21. *Let $f : (\Omega, \mathcal{F}) \rightarrow (\Omega', \mathcal{G})$ be a function, and let $\mathcal{C}$ be a generating set of $\mathcal{G}$, i.e., $\sigma(\mathcal{C}) = \mathcal{G}$. Then, f is $(\mathcal{F}, \mathcal{G})$ -measurable $\iff f^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{C}$.*

Proof. The “only if” part follows directly from the definition of measurable functions. To prove the other direction, suppose $f^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{C}$. Let

$$\mathcal{H} \triangleq \{B \subseteq \Omega' : f^{-1}(B) \in \mathcal{F}\}.$$

We claim that $\mathcal{H}$ is a σ -algebra containing $\mathcal{C}$.

To see this, note that Ω' is in $\mathcal{H}$ since $f^{-1}(\Omega') = \Omega$. Moreover, if $B \in \mathcal{H}$, then $f^{-1}(B) \in \mathcal{F}$, and by (4.1), we have

$$f^{-1}(B^c) = (f^{-1}(B))^c,$$

proving that $f^{-1}(B^c)$ is in $\mathcal{F}$, and hence, B^c is in $\mathcal{H}$. Similarly, by applying (4.2) and (4.3), we can show that $\mathcal{H}$ is closed under countable intersection and countable union. This proves that $\mathcal{H}$ is a σ -algebra.

By assumption, $\mathcal{H}$ contains all sets in $\mathcal{C}$. Hence, $\mathcal{C} \subseteq \sigma(\mathcal{C}) \subseteq \mathcal{H}$. Since $\mathcal{G}$ is the same as $\sigma(\mathcal{C})$, we get $\mathcal{G} = \sigma(\mathcal{C}) \subseteq \mathcal{H}$, i.e., every set in $\mathcal{G}$ is in $\mathcal{H}$. By the definition of $\mathcal{H}$, we obtain $f^{-1}(A) \in \mathcal{F}$ for any set A in $\mathcal{G}$. $\square$

Theorem A.22. *A continuous function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is $(\mathcal{B}(\mathbb{R}^m), \mathcal{B}(\mathbb{R}^n))$ -measurable. In particular, a continuous real-valued function $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is measurable.*

Proof. Recall that $\mathcal{B}(\mathbb{R}^n)$ is the smallest σ -algebra containing all open balls in $\mathbb{R}^n$ (see Definition 2.8). Since f is continuous, the inverse image of any open ball under f is an open set in $\mathbb{R}^m$, which belongs to $\mathcal{B}(\mathbb{R}^m)$. Hence, by Theorem A.21, f is $(\mathcal{B}(\mathbb{R}^m), \mathcal{B}(\mathbb{R}^n))$ -measurable. $\square$

Now we can deal with different kinds of operations.

Proposition A.23. *if $f, g : \Omega \rightarrow \mathbb{R}$ are $\mathcal{F}$ -measurable functions, then $f + g$, $f - g$, $c \cdot f$, f/g are measurable, assuming c constant and $g(\omega) \neq 0 \forall \omega \in \Omega$.*

Proof. See [Shu23], Chapter 4.3. $\square$

Let us define $\bar{\mathbb{R}} := \mathbb{R} \cup \{\pm\infty\}$, which is the extended real line. We can state another result involving measurability of supremum and infimum:

Theorem A.24. *If $f_i : \Omega \rightarrow \bar{\mathbb{R}}$ are measurable $\forall i = 1, 2, \dots$. Then*

$$X(\omega) := \sup_i f_i(\omega) \quad \text{and} \quad Y(\omega) := \inf_i f_i(\omega)$$

are measurable.

Proof. For a fixed $r \in \mathbb{R}$, we have

$$\{\omega : \sup_i f_i(\omega) \leq r\} = \bigcap_{i=1}^{\infty} \{\omega : f_i(\omega) \leq r\}$$

Since the right-hand side is a countable intersection of measurable sets, the left-hand side is also measurable. By Theorem A.21, $\sup_i f_i$ is measurable. We prove measurability of $\inf_i f_i$ by noting that $\inf_i f_i = -\sup_i (-f_i)$. $\square$

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The operations discussed in this section are sufficient for most applications (e.g. check if $Z = f(X, Y)$ is a random variable). In general, the cdf $F_Z(z) = P(Z \leq z)$ may not have a closed-form expression, but we can guarantee that it is well-defined because $\{\omega : Z(\omega) \leq z\}$ is a measurable set.

A.2.3 Borel-Cantelli lemmas

We use the notation $P(A_i \text{ i.o.})$ to represent the probability that event A_i happens for infinitely many indices i . The shorthand notation "i.o." stands for "infinitely often."

Theorem A.25 (Borel–Cantelli Lemma 1). *If A_i are events in a probability space $(\Omega, \mathcal{F}, P)$, for $i \geq 1$, satisfying $\sum_{i=1}^{\infty} P(A_i) < \infty$, then $P(\limsup A_i) = P(A_i \text{ i.o.}) = 0$.*

Proof. We can write the limsup of A_i as an intersection: $\limsup A_i = \bigcap_{k=1}^{\infty} \bigcup_{j=k}^{\infty} A_j$. Define the event E_k as $E_k = \bigcup_{j=k}^{\infty} A_j$. Since $E_1 \supseteq E_2 \supseteq E_3 \supseteq \dots$ is a decreasing sequence of events, the upper semi-continuity of measure implies that $P(E_k) \rightarrow 0$ as $k \rightarrow \infty$. By the sub-additivity property of measures, $P(E_k) = P\left(\bigcup_{j=k}^{\infty} A_j\right) \leq \sum_{j=k}^{\infty} P(A_j)$. Since $\sum_{j=1}^{\infty} P(A_j)$ is a convergent series, it follows that $\lim_{k \rightarrow \infty} \sum_{j=k}^{\infty} P(A_j) = 0$, implying $P(E_k) \rightarrow 0$ as $k \rightarrow \infty$. This completes the proof. $\square$

Theorem A.26 (Borel–Cantelli Lemma 2). *If $(A_i)_{i=1}^{\infty}$ is a sequence of independent events in a probability space $(\Omega, \mathcal{F}, P)$ such that $\sum_{i=1}^{\infty} P(A_i) = \infty$, then $P(\limsup A_i) = P(A_i \text{ i.o.}) = 1$.*

Proof. Consider the complement of the limsup event: $(\limsup A_i)^c = \left(\bigcap_{k=1}^{\infty} \bigcup_{j=k}^{\infty} A_j\right)^c = \bigcup_{k=1}^{\infty} \bigcap_{j=k}^{\infty} A_j^c$. We want to show that $P\left(\bigcup_{k=1}^{\infty} \bigcap_{j=k}^{\infty} A_j^c\right) = 0$. It suffices to prove that $P\left(\bigcap_{j=k}^{\infty} A_j^c\right) = 0$ for all k . We use the inequality $1 - x \leq e^{-x}$, which holds for all $x \in \mathbb{R}$. For any k , consider an integer m larger than k .

$$P\left(\bigcap_{j=k}^m A_j^c\right) = \prod_{j=k}^m P(A_j^c) = \prod_{j=k}^m (1 - P(A_j)) \leq \prod_{j=k}^m e^{-P(A_j)} = e^{-\sum_{j=k}^m P(A_j)}.$$

In the first equality above, we use the property that $A_k^c, \dots, A_m^c$ are independent events (Theorem 5.6). Since $\sum_{i=1}^{\infty} P(A_i)$ is a divergent series, we have $\sum_{j=k}^m P(A_j) \rightarrow \infty$ as $m \rightarrow \infty$. Therefore, using continuity from above, we obtain $P\left(\bigcap_{j=k}^{\infty} A_j^c\right) = 0$. Since it is true for all k , we prove $P\left(\bigcup_{k=1}^{\infty} \bigcap_{j=k}^{\infty} A_j^c\right) = 0$. $\square$

Combining the two Borel–Cantelli lemmas, we have the following zero–one law.

Theorem A.27 (Borel’s Zero–One Law). *Suppose $(A_i)_{i=1}^{\infty}$ is a sequence of independent events in a probability space $(\Omega, \mathcal{F}, P)$. Then*

$$P(A_i \text{ i.o.}) = \begin{cases} 1 & \text{if } \sum_{i=1}^{\infty} P(A_i) = \infty, \\ 0 & \text{if } \sum_{i=1}^{\infty} P(A_i) < \infty. \end{cases}$$

A.3 Lebesgue integral

Consider a sequence of random variables $X_n(\omega)$ that converge pointwise to a function $X(\omega)$. We showed that $X(\omega)$ is measurable. It is often interesting to know whether the limit of the expected values $E(X_n)$ will converge to the expected value of the limit $E(X)$. However, we cannot always exchange the order of limit and expectation. For Riemann integration, there is a helpful condition called uniform convergence. Lebesgue’s approach to integration differs. First, the integral is defined for the class of simple functions and then extended to nonnegative functions and real-valued functions.

A.3.1 Simple Functions

Definition A.28. Let $(\Omega, \mathcal{F}, \mu)$ a measure space. A function $f : \Omega \rightarrow \bar{\mathbb{R}}$ is called *simple* if its range is finite and it is $\mathcal{F}$ -measurable.

From the definition, we can write such functions as $X(\omega) = \sum_{i=1}^n a_i \mathbb{1}_{A_i}(\omega)$ with a_i values in the range of X (i.e. $\bar{\mathbb{R}}$) and $\mathbb{1}_{A_i}(\omega)$ indicator functions for the partition $A_1, \dots, A_n$ of Ω . In particular, $A_i = X^{-1}(\{a_i\})$. Note that since X is $\mathcal{F}$ -measurable, then also every element of the partition is.

Definition A.29. The Lebesgue integral of a simple function X is defined as

$$\int X d\mu := \sum_{i=1}^n a_i \mu(A_i)$$

Unless the summation includes both ∞ and $-\infty$. In that case, the integral is not defined.

These integrals have some crucial properties, often exploited to prove important results:

Theorem A.30. Let X and Y be two measurable functions. Then:

- (i) $\int \alpha X d\mu = \alpha \int X d\mu$;
- (ii) $\int (X + Y) d\mu = \int X d\mu + \int Y d\mu$;
- (iii) if $X \leq Y$, then $\int X d\mu \leq \int Y d\mu$.

In other words, the Lebesgue integral is linear and monotone.

Proof. (i) For $\alpha = 0$ it is trivially true. Assume $\alpha \neq 0$. Then,

$$\int \alpha X d\mu = \sum_{i=1}^m (\alpha a_i) \mu(A_i) = \alpha \sum_{i=1}^m a_i \mu(A_i) = \alpha \int X d\mu.$$

- (ii) Since measurability is closed under addition, $X + Y$ is measurable. If X takes m values and Y takes n , $X + Y$ cannot take more than $n \cdot m$ values. Thus, its range is still finite. We can use the partition $\{A_i \cap B_j : i = 1, \dots, m; j = 1, \dots, n\}$ to write $X + Y = \sum_{i=1}^m \sum_{j=1}^n (a_i + b_j) \mathbb{1}_{A_i \cap B_j}$. Then,

$$\sum_{i=1}^m \sum_{j=1}^n (a_i + b_j) \mu(A_i \cap B_j) = \sum_{i=1}^m \sum_{j=1}^n a_i \mu(A_i \cap B_j) + \sum_{i=1}^m \sum_{j=1}^n b_j \mu(A_i \cap B_j) = \sum_{i=1}^m a_i \mu(A_i) + \sum_{j=1}^n b_j \mu(B_j).$$

- (iii) Assume $X \leq Y$. Then $Y - X \geq 0$, and since the integral of a non-negative measurable function is non-negative, we have

$$\int (Y - X) d\mu \geq 0 \quad \Rightarrow \quad \int Y d\mu - \int X d\mu \geq 0 \quad \Rightarrow \quad \int X d\mu \leq \int Y d\mu.$$

□

A.3.2 Nonnegative Functions

To define the Lebesgue integral of a nonnegative function $X : \mathbb{R} \rightarrow \mathbb{R}$, we first approximate it from below by simple functions and then take the supremum of their integrals.

Definition A.31. For a nonnegative measurable function X , we define the *Lebesgue integral* of X by

$$\int X d\mu := \sup \left\{ \int f d\mu : f \text{ is simple, } 0 \leq f \leq X \right\}.$$

If we define the set $S(X) := \{f : f \text{ is simple, } 0 \leq f \leq X\}$, then the Lebesgue integral can be expressed as

$$\int X d\mu = \sup \left\{ \int f d\mu : f \in S(X) \right\}.$$

If the set $\{\int f d\mu : f \in S(X)\}$ is unbounded, then $\int X d\mu = \infty$. Furthermore, if $X(\omega)$ is equal to ∞ for ω in a set A with positive measure, then the integral $\int X d\mu$ is also equal to ∞ .

We note that the set $S(X)$ is not empty. One way to approximate a nonnegative measurable function X by simple functions is as follows. For any positive integer n , we define a simple function $X^{(n)}(\omega)$ by

$$X^{(n)}(\omega) = \begin{cases} \frac{k}{2^n} & \text{if } \frac{k}{2^n} \leq X(\omega) < \frac{k+1}{2^n} \text{ for some } k = 0, 1, 2, \dots, n2^n - 1 \\ n & \text{if } n \leq X(\omega). \end{cases}$$

Moreover, monotonicity is a direct consequence of the definition. Linearity comes from the next result. At this point, we can state an important result that summarizes the behavior of integrals for non-decreasing sequences of functions, which is a fundamental result both in measure theory and in probability theory.

Theorem A.32 (Monotone Convergence Theorem). *If $0 \leq X_1 \leq X_2 \leq X_3 \leq \dots$ is a sequence of non-decreasing, nonnegative measurable functions converging to X from below, then*

$$\int X_n d\mu \rightarrow \int X d\mu \quad \text{as } n \rightarrow \infty.$$

Proof. [Shu23], Chapter 6.2. □

A.3.3 Real-Valued Functions

Definition A.33. For a real-valued measurable function X , we let

$$X^+ := \max(X, 0) \text{ and } X^- := -\min(X, 0)$$

be the positive and negative part of X , respectively.

Both X^+ and X^- are measurable because the functions $\max(x, 0)$ and $-\min(x, 0)$ are continuous. Moreover, we have

$$X = X^+ - X^- \text{ and } |X| = X^+ + X^-.$$

Using these concepts, we can define the Lebesgue integral of a real-valued function by reducing the definition to the nonnegative case.

Definition A.34 (Real Lebesgue Integral). We define the *Lebesgue integral* of a measurable $\bar{\mathbb{R}}$ -valued function X on a measure space $(\Omega, \mathcal{F}, \mu)$ by

$$\int X \, d\mu := \int X^+ \, d\mu - \int X^- \, d\mu.$$

It is well-defined unless we have $\infty - \infty$ on the right-hand side. When $\int X^+ \, d\mu$ and $\int X^- \, d\mu$ are both finite, we say that X is μ -integrable and write $X \in L^1(\mu)$. It is possible to show that X is integrable if and only if $\int |X| \, d\mu < \infty$.

Linearity also holds in the case of real measurable functions.

Now that we have developed these tools, we can finally define the mathematical expectation of a random variable.

Definition A.35. The *expectation* of a random variable X is defined by

$$E[X] := \int_{\Omega} X \, dP.$$

A.4 Convergence

In this section, we state (without proofs) some useful results to prove theorems in other chapters.

Theorem A.36 (Triangle Inequality for Real Integral).

$$\left| \int X \, d\mu \right| \leq \int |X| \, d\mu \quad \text{for } X \in L^1(\mu).$$

The triangle inequality implies that the L^1 norm satisfies two of the defining properties of a norm function, which are:

$$\begin{aligned} \|X + Y\|_1 &\leq \|X\|_1 + \|Y\|_1 \\ \|\alpha X\|_1 &= |\alpha| \cdot \|X\|_1 \quad \text{for any scalar } \alpha. \end{aligned}$$

Theorem A.37 (Fatou's Lemma). Let X_n , $n = 1, 2, 3, \dots$, be a sequence of measurable functions defined on a measure space $(\Omega, \mathcal{F}, \mu)$, such that each of them is nonnegative almost everywhere. Then

$$\int \liminf_{n \rightarrow \infty} X_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int X_n \, d\mu.$$

Theorem A.38 (Dominated Convergence Theorem). *For $n = 1, 2, 3, \dots$, let $X_n(\omega)$ be $\mathbb{R}$ -valued measurable functions defined on a measure space $(\Omega, \mathcal{F}, \mu)$, converging almost everywhere to $X(\omega)$, i.e. $X_n \rightarrow X$ a.e..*

If there exists a nonnegative integrable function Y such that $|X_n| \leq Y$ a.e. $\forall n$, then

(i) *X is Lebesgue integrable, i.e., $X \in L^1(\mu)$;*

(ii)

$$\lim_{n \rightarrow \infty} \int X_n d\mu = \int X d\mu = \int \lim_{n \rightarrow \infty} X_n d\mu.$$

Moreover, we have L^1 convergence: $\int |X_n - X| d\mu \rightarrow 0$ as $n \rightarrow \infty$.

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